ANNEXURE A – INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization

A mobile manufacturer observes lag while switching between apps. Analyze the role of L1,

L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and

improve throughput.

Problem: Lag occurs while switching between applications.

Analysis:

- **L1 Cache:** Very small and extremely fast; stores frequently used instructions and data.
- **L2 Cache:** Larger than L1 and provides a balance between speed and capacity.
- **L3 Cache:** Larger shared cache that reduces frequent access to slower DRAM.
- **DRAM:** Provides large storage capacity but has much higher latency than cache memory.

Proposed Solution:
Use a **hierarchical memory design** with larger L2/L3 caches and optimized DRAM. Frequently accessed application data should remain in L1/L2 cache, while shared data can be stored in L3. This reduces DRAM accesses, decreases app-switching latency, and improves overall throughput.

2. Cloud Server Memory Bottleneck

A cloud service provider experiences high latency in database queries. Compare cache

hierarchy vs virtual memory paging and recommend improvements using appropriate

memory technologies.

Problem: Database queries experience high latency.

Analysis:

- **Cache hierarchy:** L1/L2/L3 caches provide very fast access to frequently used database instructions and data.
- **Virtual memory paging:** Uses secondary storage when physical RAM is insufficient, but SSD/storage access is much slower than RAM.
- Excessive paging causes **page faults**, increasing query response time.

Recommended Solution:
Increase **DRAM capacity**, optimize L3 cache usage, and use **NVMe SSDs** for faster virtual-memory operations. Database caching should also be used to keep frequently accessed records in RAM, reducing paging and improving query performance.

3. Autonomous Vehicle Data Processing

An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and

MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

Problem: Sensor data must be processed in real time with minimum delay.

Memory Comparison:

- **SRAM:** Extremely fast and low latency but expensive and limited in capacity.
- **DRAM:** High capacity and good bandwidth, but slower than SRAM.
- **MRAM:** Non-volatile, fast, and consumes less standby power than conventional memory.

Proposed Memory Hierarchy:
Use **SRAM as L1/L2 high-speed working memory**, **DRAM as the main memory for large sensor datasets**, and **MRAM for persistent data, configuration, and frequently required information**. This structure provides high bandwidth, low latency, and reliable real-time processing.

4. AI Inference Edge Device

An edge AI device needs fast model loading with low power consumption. Compare NAND

Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

Problem: The device needs fast AI model loading while consuming low power.

Memory Comparison:

- **NAND Flash:** High capacity, non-volatile, low cost, but slower for direct computation.
- **DRAM:** Fast and suitable for active AI model execution but consumes more power.
- **RRAM:** Non-volatile, fast, low-power memory that can support AI-oriented computing.

Recommended Solution:
Store the AI model permanently in **NAND Flash**, load frequently used model parameters into **RRAM**, and use **DRAM as the working memory** during inference. This hierarchy reduces model-loading time, improves inference speed, and lowers power consumption.

5. High-Performance Gaming Console

A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs

DRAM throughput and propose memory hierarchy enhancements.

Problem: Frame drops occur when loading large game scenes.

Analysis:

- **L3 Cache:** Provides low-latency access to frequently reused game data but has limited capacity.
- **DRAM:** Provides much larger capacity and high bandwidth for textures, game assets, and scene data, but has higher latency than cache.
- Frame drops can occur when large amounts of data must repeatedly move between DRAM and the processor.

Proposed Solution:
Increase the effectiveness of the **L3 cache** and use **high-bandwidth DRAM**.

Frequently accessed game data such as CPU instructions and important scene information should remain in cache, while large textures and assets should be efficiently streamed from DRAM. Better cache management and higher DRAM bandwidth can reduce loading delays and provide smoother frame rates.